

Ashmit Cajla

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Data Engineering Intern candidate · available 6 months, Jan–Jul 2027

EDUCATION

Thapar Institute of Engineering and Technology

B.Tech., Electronics and Computer Engineering; CGPA 7.62/10

Patiala, India

Aug 2023 – Jul 2027

Indian Institute of Technology Madras (online degree, concurrent)

B.S., Data Science and Applications

Chennai, India

Aug 2023 – Aug 2027

Coursework: Database Management Systems, Data Structures & Algorithms, Operating Systems, Computer Networks, ML

EXPERIENCE

Data Scientist & Portal Developer

Apr 2025 – Jan 2026

Tel Aviv University (agri-IoT research data platform)

Remote

- Built and operated the ingestion and query pipeline (**Python, FastAPI, GCP** + self-managed nodes) for **2M+ sensor events/day** from 1,500+ sensors across 300 farms, storing terabytes of time-series data.
- Added a **data-quality layer** on incoming records: anomaly and duplicate detection, structured failure logging and automatic retries, raising platform availability to **99.4%** and cutting infrastructure cost by **30%**.
- Served validated data to downstream monitoring tools through internal REST APIs; investigated unexpected patterns flagged by the checks with domain researchers and refined the detection rules.
- Shipped through Git pull requests and code review in a 5-person team of engineers and researchers.

Student Researcher, Samsung PRISM '24

Jun 2024 – Mar 2025

Samsung R&D Institute India

Remote

- Built a generative-AI detector for visible and invisible image watermarks (**95%+ precision**).

UAV Engineer, Autonomous Systems

Jun 2025 – Jul 2025

Indian Army, 10 Corps Zonal Workshop

Suratgarh, India

- Built and evaluated autonomous decision-making modules for UAV systems with **sub-5-second** response times.

PROJECTS

VenturesLab – 12-agent LLM analysis platform | github.com/AshmitCajla/VenturesLab

2026

- FastAPI service orchestrating 12 LLM agents (LangGraph) with **RAG** over uploaded PDFs: chunking, bge-small embeddings, similarity search in **Qdrant** or in-memory, and prompts grounded in the retrieved passages.
- Wrote 20+ automated output checks (range, ordering and consistency rules) that trigger a corrective retry, plus failure isolation per step; 14 tests and Docker builds run in GitHub Actions CI. Built with an AI coding assistant; every change reviewed and tested.
- Long analyses run as background jobs with per-step progress polling, so multi-minute runs survive request timeouts; LLM provider fallback and SSRF-safe URL fetching.

ResearchOS – literature knowledge graph | github.com/AshmitCajla/research_OS

2026

- Pulls up to 50 papers from two APIs, deduplicates and ranks them by embedding similarity, extracts reported results from PDFs into structured records, and canonicalises names into a model×dataset graph.
- Exponential-backoff retries on rate limits, arXiv fallback when throttled, parallel full-text reads.

AI-Driven Next-Gen Firewall

Dec 2025

- Real-time streaming pipeline: Suricata signature inspection → **Kafka** event stream → LightGBM flow classifier → SOAR-style engine that automatically blocks malicious traffic. Winning team, Smart India Hackathon '25.
- 99.16% accuracy, 99.10% recall on held-out CICIDS2017; integration-tested with attack simulations.

SKILLS

Languages: Python, SQL, Java, C++ , JavaScript

Data: data pipelines, data-quality checks, time-series data, Kafka, REST APIs, Spark (coursework)

LLM / AI: prompting, RAG, embeddings, vector search (Qdrant), LLM output evaluation; daily use of AI coding assistants

Tools: Git, Linux, Docker, GCP, AWS, GitHub Actions, FastAPI

PUBLICATIONS & IP

- Securing IoT Devices using a Multi-Layer IDS Framework.** IEEE IC3 2026, accepted and presented. Co-author.
- Defending Large Audio-Language Models against Audio-Based Prompt Injections via Defensive Fine-Tuning.** IEEE IC3 2026. Co-author.
- Act-Reco: Real-time activity recognition framework based on deep learning.** *Evolving Systems* (Springer), 2026. doi.org/10.1007/s12530-025-09779-6.
- UAV Detection and Neutralisation Device.** Registered Design No. 486614-001 (Jan 2026).

LEADERSHIP & AWARDS

Winning team, SIH '24 (3D commando tracker) · President, GDG Thapar · President, Student Alumni Interaction Cell